Code No: 157AA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
AD-HOC AND SENSOR NETWORKS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Distinguish a Mobile ad hoc Network (Manet) from other types of networks.
b) List and explain the important characteristics of a Manet. [7+8]

2.a) Explain the advantages and disadvantages of DSDV algorithm.
b) What are temporal and spatial resolutions in DREAM location service? [7+8]

3.a) Illustrate the broadcast storm problem in ad hoc networks.
b) Explain the taxonomy of broadcast routing protocols. [7+8]

4.a) Explain the pros and cons of tree-based and mesh-based multicast solutions.
b) Explain the mechanism in any hybrid multicast protocol for Manets. [7+8]

5.a) What is the significance of geocast routing in Manet applications? Give the taxonomy of geocast protocols.
b) Illustrate the concept in Mesh-based Geocast Routing (MGR) protocol. [7+8]

6.a) List and brief the reasons for poor performance of TCP over ad hoc network.
b) Explain how the problem of TCP unfairness problem is addressed. [7+8]

7.a) Draw the diagram for functional components of a typical wireless sensor node.
b) Briefly write about the spectrum of applications of wireless sensor networks. [7+8]

8. Write about the following in wireless sensor networks:
a) Distributed query processing
b) In-network processing. [7+8]

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